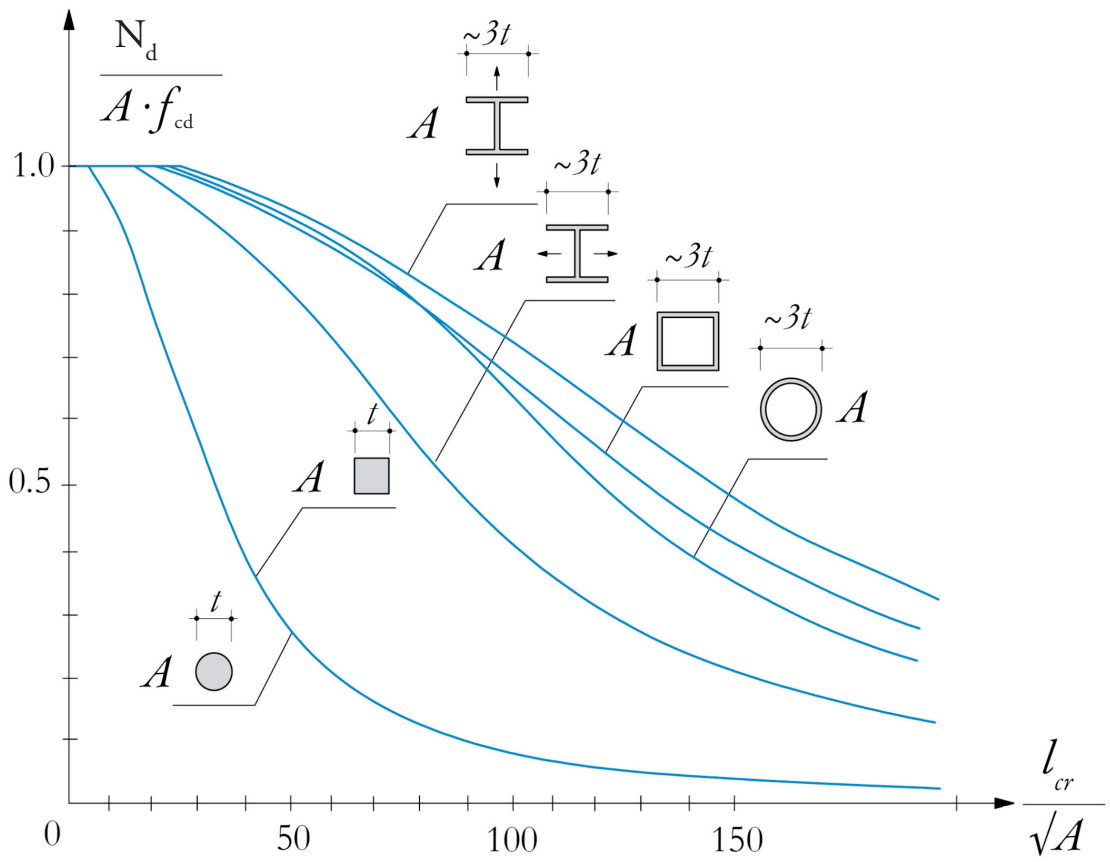
Task 4 Stability against buckling

Given is a 2.4 m long column made of steel S355. It is designed as a round solid profile with a cross-sectional area of 400 mm². It is subjected to a force N_d=54 kN, and has a hinged support at the top and bottom.

a) Check the buckling behaviour of the column. Use the relevant buckling curve to check its stability and draw the corresponding values in the diagram.

b) In general, how can buckling be prevented?



buckling behaviour for different steel profiles

a)

b)